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MEDLY

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Project Guide

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Contents

- Introduction
- Objectives
- Scope
- Features / Functionality
- Project Outcomes
- Technology Stack
- Results

1. Introduction

- **Problem Identified :**

Managing daily medication is challenging due to missed doses, unsafe drug combinations, and difficulty identifying symptoms. In emergencies, finding nearby pharmacies is also difficult, and no lightweight offline app currently combines all these features in one place.

- **Solution Proposed :**

Medly combines six essential health tools into one offline-first Flutter app. It enables fuzzy medicine search, dose reminders, drug interaction checks, symptom analysis, and a pharmacy finder with real-time details when online.

SDG (Sustainable Development Goals)

SDG 3:
Good Health and Well-being
Ensures better health through medicine management, reminders, drug safety, and symptom analysis

2. Objectives

- To store and manage medicine information in an organized way.
- To help users find medicines and get suggestions based on symptoms.
- To provide reminders for taking medicines and track missed doses.
- To ensure safe usage by checking drug interactions and finding nearby pharmacies.

3. Scope

1. The application can be applied in the healthcare and medical domain to help users quickly access medicine information.It can be used in pharmacies and medical stores to identify medicines
2. The system can help patients and general users find nearby pharmacies where a particular medicine is available.It can be useful for doctors and healthcare workers to quickly check medicine details.

4. Feature /Functionality

Feature 1: Intelligent Medicine Search & Information Display
The application allows users to search medicines using fuzzy matching and view key details like usage, dosage, and side effects from a local database.

Feature 2: Medication Management & Safety Alerts
The app enables reminder scheduling, dose tracking, and alerts for missed doses and potential drug interactions.

Feature 3: Symptom Analysis & Pharmacy Locator
The system analyzes symptoms to suggest possible conditions and helps users locate nearby pharmacies using location data.

5. Outcome of Project

1. User can search and identify medicines efficiently using intelligent fuzzy search.
2. User can view detailed and accurate information about medicines including usage, dosage, and side effects.
3. User can set reminders, track medication intake, and monitor adherence.
4. User receives alerts for missed doses and potential drug interactions.
5. User can analyze symptoms and get suggestions for possible conditions and treatments.
6. User can locate nearby pharmacies using built-in location and distance calculation methods.

6. Technology Stack

1. Frontend (GUI)

- Framework/Language: Flutter (Dart) for cross-platform Android and iOS
- State Management: Provider with structured service layer
- UI: Material Design 3 with clean and responsive components
- Features: Local notifications for reminders, URL launcher for maps and calls

2. API & Platform:-

- API: Geoapify Places API for pharmacy search
- Response: GeoJSON with location and contact details
- Platform: Android and iOS support with offline-first functionality

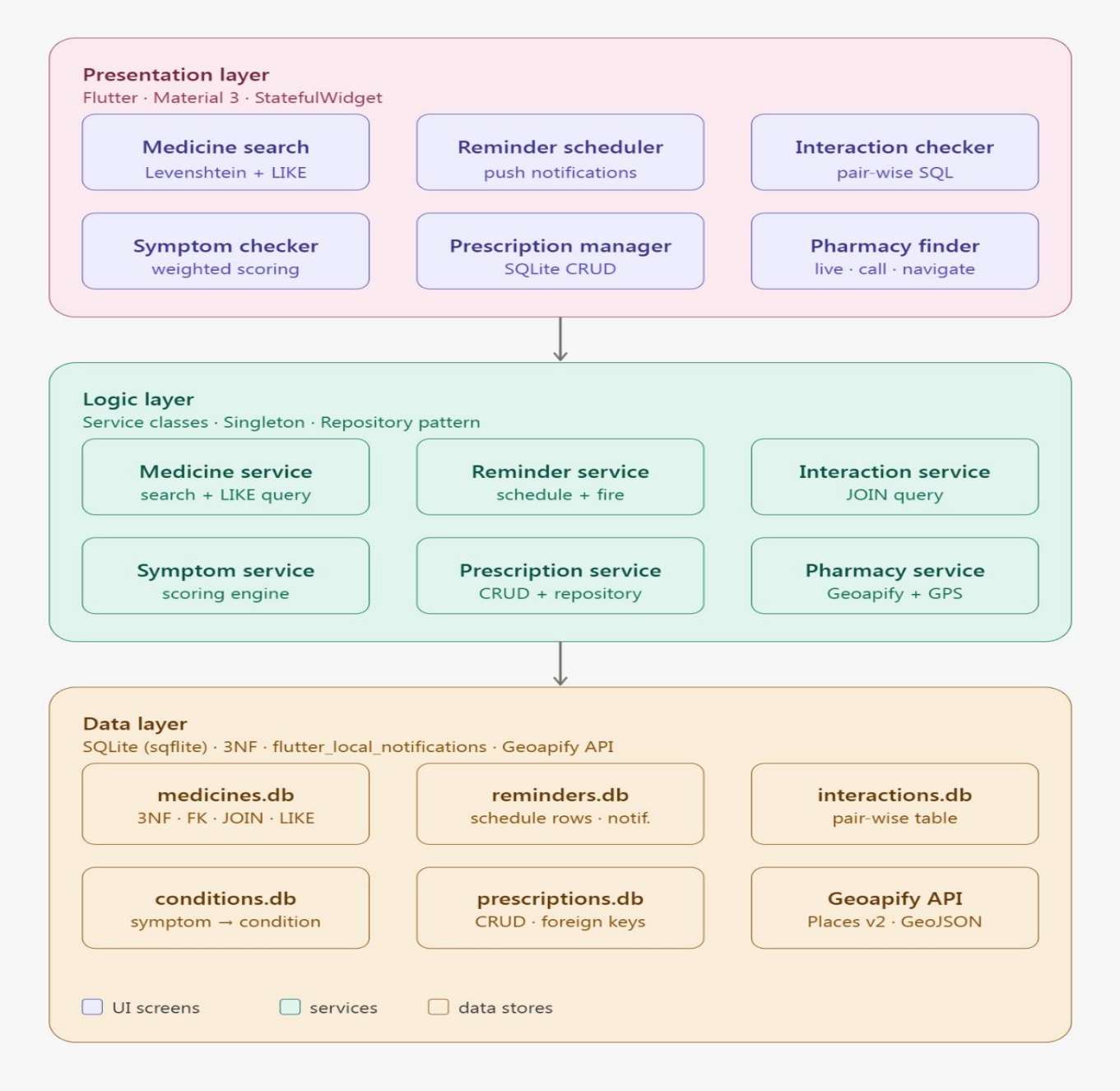
3.Database (Local Storage):-

- Database: SQLite using sqflite
- Design: 3NF normalized relational schema
- Queries: Foreign keys, JOIN operations, LIKE search
- Storage: Separate databases for medicines, reminders, interactions, conditions

4. Back End (Logic & Services)

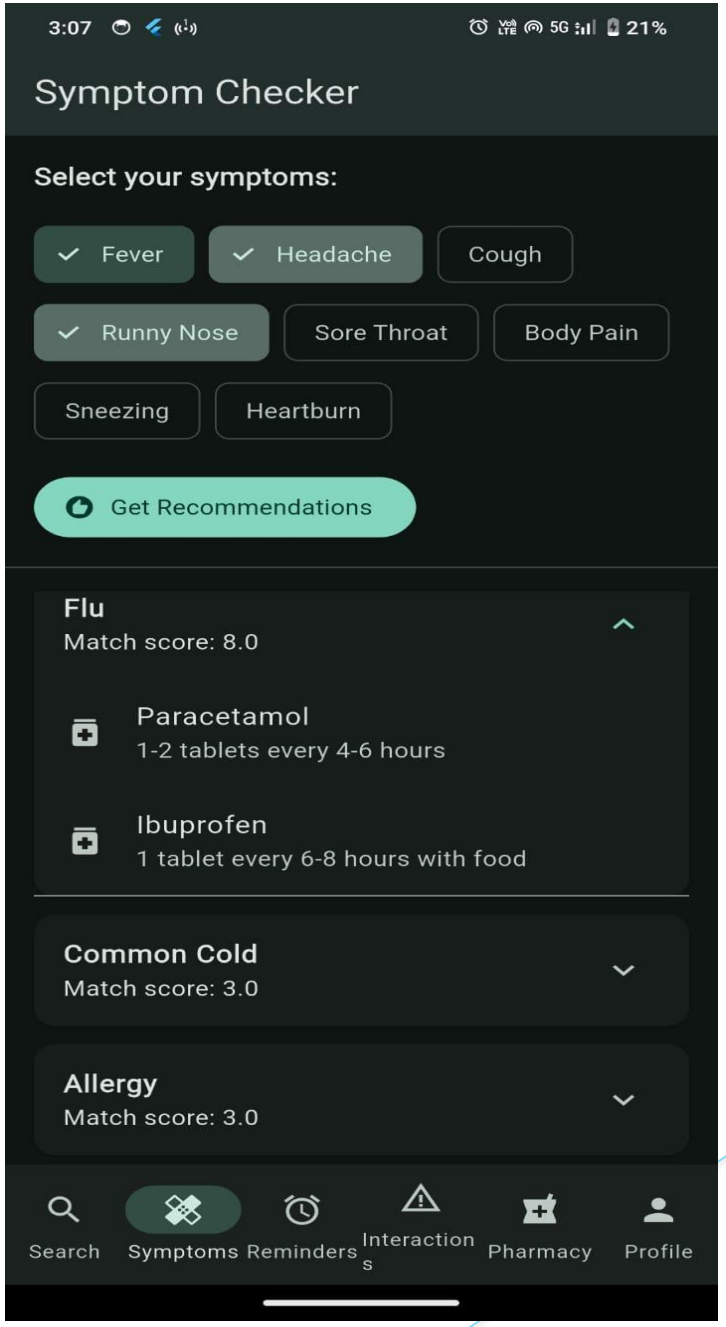
- Architecture: Service Layer with Singleton and Repository pattern
- Logic: Levenshtein algorithm for fuzzy search
- Distance: Haversine formula for pharmacy sorting
- Engine: Rule-based symptom analysis with weighted scoring
- Safety: Drug interaction checking using pair-wise logic
- Integration: HTTP for API handling, Geolocator for GPS

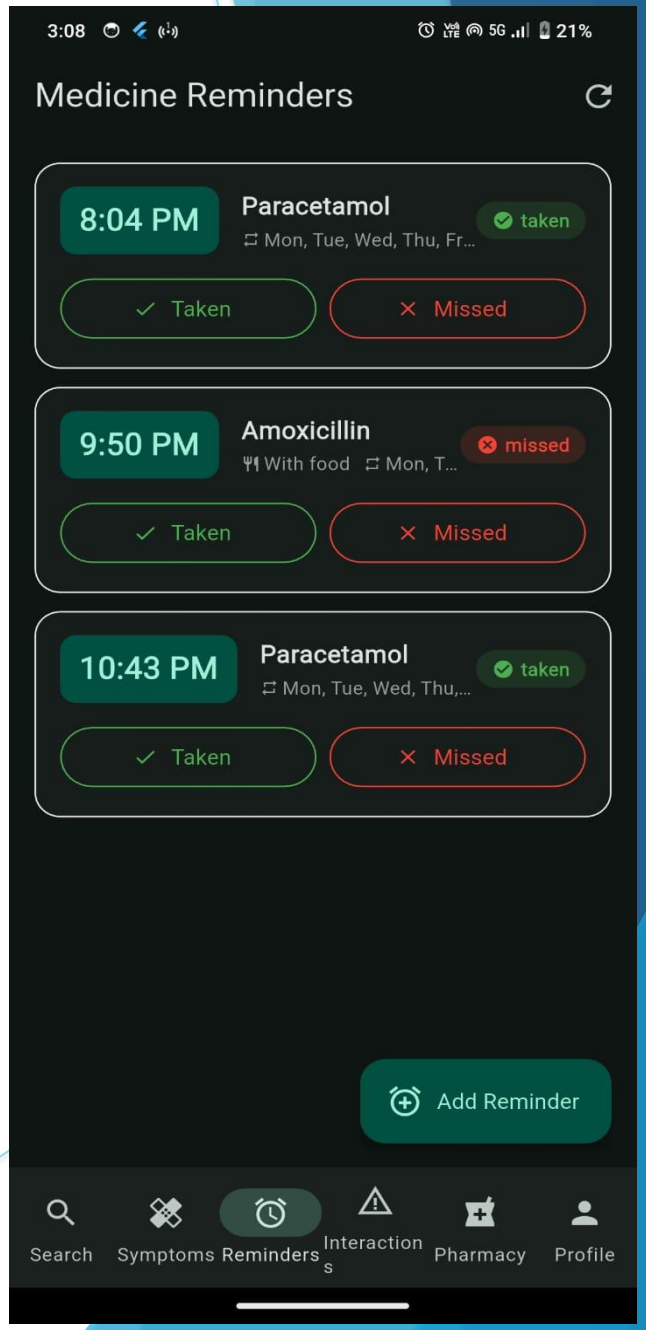
8. Block Diagram (if applicable)



10.Result

3:07
21%
Medly
Search medicines, compositions, or symptoms
Search medicines, eg. Paracetamol...
All Medicines
Paracetamol
Acetaminophen 500mg
tablet
Analgesic / Antipyretic
Ibuprofen
Ibuprofen 400mg
tablet
NSAID / Anti-inflammatory
Amoxicillin
Amoxicillin 500mg
capsule
Antibiotic (Penicillin group)
Cetirizine
Cetirizine HCl 10mg
tablet
Antihistamine
Omeprazole
Omeprazole 20mg
capsule
Proton Pump Inhibitor (PPI)
Search
Symptoms
Reminders
Interactions
Pharmacy
Profile


3:07
21%
Symptom Checker
Select your symptoms:
Fever
Headache
Cough
Runny Nose
Sore Throat
Body Pain
Sneezing
Heartburn
Get Recommendations
Flu
Match score: 8.0
Paracetamol
1-2 tablets every 4-6 hours
Ibuprofen
1 tablet every 6-8 hours with food
Common Cold
Match score: 3.0
Allergy
Match score: 3.0
Search
Symptoms
Reminders
Interactions
Pharmacy
Profile


3:08
21%
Medicine Reminders
8:04 PM
Paracetamol
taken
Mon, Tue, Wed, Thu, Fr...
Taken
Missed
9:50 PM
Amoxicillin
missed
With food
Mon, T...
Taken
Missed
10:43 PM
Paracetamol
taken
Mon, Tue, Wed, Thu,...
Taken
Missed
Add Reminder
Search
Symptoms
Reminders
Interactions
Pharmacy
Profile

3:08 5G 21%

Nearby Pharmacies

Find Real Pharmacies Near You

Lat: 19.1848, Lon: 72.9916

Search Again

4 pharmacies found nearby

Pharmacy Nearest

1.15 km

Pharmacy, 123, SC Bose Road, Jambli Naka, Thane - 400605, Maharashtra, India

Navigate

2 Vijay Medical & General Stores

2.08 km

Vijay Medical & General Stores, Sane Guruji Road, Naupada, Thane - 400601, Maharashtra, India

9757432665 | Navigate

3 BRIGHT HOMOEOPATHIC PHARMAC

2.15 km

BRIGHT HOMOEOPATHIC PHARMAC, Mavli Mandal Marg, Uthalsar, Thane - 400601, Maharas...

Search | Symptoms | Reminders | Interactions | Pharmacy | Profile

3:10 5G 21%

My Health Profile

Low Stock: Amoxicillin

2 tablets remaining — time to refill!

Basic Information

Full Name: bhumika

Age: 18

Gender: Female

Weight (kg): 35

Height (cm): 155

Blood Group: AB+

Medical Conditions

Blood Pressure

Normal | Low BP | High BP

Diabetes

None | Pre-diabeti | Type 2

+ Track Medicine

Search | Symptoms | Reminders | Interactions | Pharmacy | Profile

3:10 5G 21%

My Health Profile

Type 2

Pregnant

Allergies

Add allergy (e.g. Penicillin)

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

hypertension ×

Medicine Stock Tracker

Amoxicillin

+ Track Medicine

Search | Symptoms | Reminders | Interactions | Pharmacy | Profile

3:10 21%

My Health Profile

Allergies

Add allergy (e.g. Penicillin)

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

hypertension ×

Medicine Stock Tracker

Amoxicillin 2

2 of 10 left Exp: 2026-09-30

+ Track Medicine

Search Symptoms Reminders Interactions Pharmacy Profile

3:10 21%

My Health Profile

Allergies

Add allergy (e.g. Penicillin)

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

Track Medicine Stock

Medicine

Aspirin

Total Count

20

Daily Dose

1

Alert when below (tablets)

5

Expires: 2026-10-18

Add to Tracker

3:10 21%

My Health Profile

penicillin ×

Chronic Conditions

Add condition (e.g. Hypertens...

hypertension ×

Medicine Stock Tracker

Amoxicillin 2

2 of 10 left Exp: 2026-09-30

Aspirin 20

20 of 20 left Exp: 2026-10-18

+ Track Medicine

Search Symptoms Reminders Interactions Pharmacy Profile

Thank You...!!